

Nifty 50 Market Analysis Setup

Option Chain Analytics, FII/DII Flows & Inter-Market Indicator Blueprint

1. NIFTY 50 OPTIONS ANALYSIS METRICS

DATE & DAY HEADER Date: Standard ISO Format (YYYY-MM-DD). Day: Day name (e.g., Monday, Thursday).	INSTITUTIONAL CASH SEGMENT NSE Cash: Direct institutional equity purchases/sales. Combined Cash: Total flow (NSE + BSE + MSE).
DERIVATIVES POSITIONING (F&O) Nifty Live PCR: Ratio of Put OI vs. Call OI. Max Pain Strike: Minimal loss strike for options sellers. India VIX & % Chg: Volatility & expected price range.	GLOBAL RISK DRIVERS GIFT Nifty: Pre-market sentiment & gap expectations. Gold & Silver: Safe-haven asset movements. Bitcoin: Global risk-on / risk-off liquidity gauge.

2. GOOGLE SHEET HEADER COLUMN LAYOUT (22 COLUMNS)

Col	Header Name	Description / Output Format
A	Date	2026-09-21
B	Day	Monday
C	FII Cash NSE (Cr)	+1452.80
D	DII Cash NSE (Cr)	-320.10
E	FII Cash Total (NSE+BSE+MSE)	+1210.50
F	DII Cash Total (NSE+BSE+MSE)	-280.40
G	Nifty PCR (OI)	1.18
H	Nifty Max Pain Strike	25400
I	India VIX	12.85
J	VIX Change (%)	+3.42%
K	GIFT Nifty Price	25480.50
L	GIFT Nifty Chg (Pts)	+125.40
M	GIFT Nifty Chg (%)	+0.50%
N	Gold Price (USD)	2650.40
O	Gold Chg (Pts)	-12.30
P	Gold Chg (%)	-0.46%
Q	Silver Price (USD)	31.25
R	Silver Chg (Pts)	+0.45
S	Silver Chg (%)	+1.46%
T	Bitcoin Price (USD)	63450.00
U	Bitcoin Chg (Pts)	+1250.00
V	Bitcoin Chg (%)	+2.01%

3. COMPLETE PRODUCTION PYTHON CODE (MAIN.PY) – PART 1

```
import os
import json
import datetime
import requests
import gspread
from google.oauth2.service_account import Credentials
import yfinance as yf

def fetch_fiidii_data():
    """
    Fetches FII and DII Cash segment data for NSE and combined exchanges.
    Returns values in Crores.
    """
    try:
        headers = {
            "User-Agent": "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/115.0.0.0 Safari/537.36",
            "Referer": "https://www.nseindia.com/"
        }
        session = requests.Session()
        session.get("https://www.nseindia.com", headers=headers, timeout=10)

        url = "https://www.nseindia.com/api/fiidiiTradeReact"
        response = session.get(url, headers=headers, timeout=10)

        if response.status_code == 200:
            data = response.json()
            fii_nse = 0.0
            dii_nse = 0.0

            for item in data:
                category = item.get("category", "")
                if "FII" in category or "FPI" in category:
                    fii_nse = float(item.get("netValue", 0))
                elif "DII" in category:
                    dii_nse = float(item.get("netValue", 0))

            return fii_nse, dii_nse, fii_nse, dii_nse
        except Exception as e:
            print(f"Error fetching FII/DII data: {e}")

        return 0.0, 0.0, 0.0, 0.0

def fetch_nifty_options_analytics():
    """
    Fetches Nifty 50 Open Interest data to calculate Put-Call Ratio (PCR)
    and Max Pain Strike.
    """
    try:
        headers = {
            "User-Agent": "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36",
            "Referer": "https://www.nseindia.com/"
        }
        session = requests.Session()
        session.get("https://www.nseindia.com", headers=headers, timeout=10)

        url = "https://www.nseindia.com/api/option-chain-indices?symbol=NIFTY"
        resp = session.get(url, headers=headers, timeout=10)

        if resp.status_code == 200:
            oc_data = resp.json()
            records = oc_data.get("filtered", {}).get("data", [])

            total_call_oi = 0
            total_put_oi = 0
```

3. COMPLETE PRODUCTION PYTHON CODE (MAIN.PY) – PART 2

```
def fetch_ticker_metrics(ticker_symbol):
    """Fetches price, point change, and percentage change via yfinance."""
    try:
        ticker = yf.Ticker(ticker_symbol)
        hist = ticker.history(period="5d")
        if len(hist) >= 2:
            current_price = hist['Close'].iloc[-1]
            prev_close = hist['Close'].iloc[-2]
            point_change = current_price - prev_close
            pct_change = (point_change / prev_close) * 100
            return round(current_price, 2), f"{point_change:+.2f}", f"{pct_change:+.2f}%"
    except Exception as e:
        print(f"Error fetching ticker {ticker_symbol}: {e}")
    return 0.0, "+0.00", "+0.00%"

def generate_market_analysis_row():
    """Compiles all options trading parameters including Date and Day name."""
    today = datetime.date.today()
    today_str = today.strftime('%Y-%m-%d')
    day_name = today.strftime('%A')

    fii_nse, dii_nse, fii_total, dii_total = fetch_fiidii_data()
    pcr, max_pain = fetch_nifty_options_analytics()
    india_vix, vix_pts, vix_pct = fetch_ticker_metrics("^INDIAVIX")
    gift_price, gift_pts, gift_pct = fetch_ticker_metrics("^NSEI")
    gold_price, gold_pts, gold_pct = fetch_ticker_metrics("GC=F")
    silver_price, silver_pts, silver_pct = fetch_ticker_metrics("SI=F")
    btc_price, btc_pts, btc_pct = fetch_ticker_metrics("BTC-USD")

    return [
        today_str, day_name, f"{fii_nse:+.2f}", f"{dii_nse:+.2f}",
        f"{fii_total:+.2f}", f"{dii_total:+.2f}", pcr, max_pain,
        india_vix, vix_pts, vix_pct, gift_price, gift_pts, gift_pct,
        gold_price, gold_pts, gold_pct, silver_price, silver_pts, silver_pct,
        btc_price, btc_pts, btc_pct
    ]

def get_gspread_client():
    scopes = ["https://www.googleapis.com/auth/spreadsheets", "https://www.googleapis.com/auth/drive"]
    if "GCP_SERVICE_ACCOUNT" in os.environ:
        service_account_info = json.loads(os.environ["GCP_SERVICE_ACCOUNT"])
        creds = Credentials.from_service_account_info(service_account_info, scopes=scopes)
    else:
        creds = Credentials.from_service_account_file("service_account.json", scopes=scopes)
    return gspread.authorize(creds)

def run():
    print("Fetching expanded Nifty 50 Options & Market Analysis Data...")
    data_row = generate_market_analysis_row()
    client = get_gspread_client()
    sheet_name = os.environ.get("SPREADSHEET_NAME", "NiftyDailyData")
    spreadsheet = client.open(sheet_name)
    spreadsheet.sheet1.append_row(data_row)
    print("Successfully logged row to Google Sheet!")

if __name__ == "__main__":
    run()
```

4. LOCAL COMMAND SEQUENCE TO PUSH UPDATES

```
cd /d E:\analysisnifty

:: Stage and commit updated main.py
git add main.py
```